The Bayes Filter: Predict, Then Update

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1. Overview

> **Key point:** The Bayes filter updates a robot's belief one step at a time: a prediction step pushes the belief through the motion model after each command, and a correction step multiplies it by the measurement probability after each reading. The robot never needs to store its history.

So far the robot keeps a belief: a probability for every possible state, given every command and every reading since it started. That list is what still breaks. It grows by one command and one reading at every step: a robot that reads its sensor 10 times a second holds 36,000 readings after an hour, and by its definition the belief must be recomputed from all of them. This Note adds the Bayes filter, which keeps only the last belief. It answers the question: **how can a robot keep its belief up to date at every step, using only the newest command and the newest reading?**

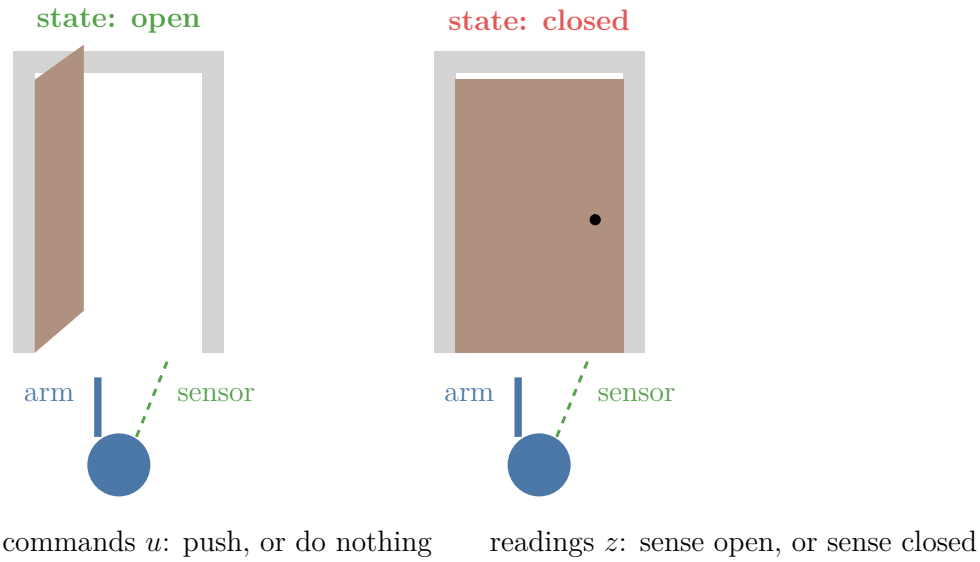


Figure 1: Thrun's door world: a door that is open (left) or closed (right); a robot in front of it with an arm that can push the door and a sensor that reads open or closed

We leave the hallway of belief for Thrun's door world (Figure 1): 2 states instead of 10 cells, so every sum can be done by hand (PR §2.4.2). The grid filters bring the filter back to the hallway. This Note covers:

- one reading at a time, with Bayes' theorem that remembers earlier readings (Section 3);
- the filter itself, a prediction step and a correction step, worked on the door (Section 4);
- why the two steps give the exact belief, and which three assumptions that needs (Section 5);
- the same two steps on a continuous position, where sums become integrals (Section 6);
- why the belief is all the robot needs to keep (Section 7), and which famous filters are this one in disguise (Section 8).

2. The door world

> **Key point:** One state (the door, open or closed), two commands (push, do nothing), one noisy sensor. Its two probability tables are all the filter needs.

The **state** x_t is the door at time t: open or closed. The robot does not know it. The two tables of belief describe the robot (PR §2.4.2); with them and its last belief, the filter never needs the list of past readings.

The measurement probability $p(z_t \mid x_t)$: the sensor reads "open" or "closed", and is sometimes wrong.

True door	p(sense open)	p(sense closed)
open	0.6	0.4
closed	0.2	0.8

These are the same numbers as the door sensor of the hallway.

The state transition probability $p(x_t \mid u_t, x_{t-1})$: the robot can push the door or do nothing.

Command	Door before	p(open after)	p(closed after)
push	open	1	0
push	closed	0.8	0.2
do nothing	open	1	0
do nothing	closed	0	1

A push on a closed door fails one time in five, because the door is heavy. Doing nothing changes nothing.

The start. The robot has no idea, so its first belief is:

$$\text{bel}(x_0 = \text{open}) = 0.5$$

$$\text{bel}(x_0 = \text{closed}) = 0.5$$

The run. At $t = 1$ the robot does nothing and senses "open". At $t = 2$ it pushes and senses "open" again. We want its belief after each step.

3. One reading at a time: Bayes' theorem with past readings

> **Key point:** Bayes' theorem turns "how likely is this reading in each state" into "how likely is each state given the reading". Applied again with the first answer as the new prior, it adds a second reading, provided the readings are independent once the state is known.

3.1 Why the robot needs Bayes' theorem

First the readings alone: can the robot take in a new reading without going back to the old ones? For now, nobody touches the door; the robot only reads. Its table tells it $p(z \mid x)$: how often the sensor reads "open" for each kind of door. The robot needs the other direction, $p(x \mid z)$: given the reading, how likely is each kind of door? Bayes' theorem (G-269) turns one into the other:

$$p(x \mid z) = \frac{p(z \mid x)\ p(x)}{p(z)}$$

First reading, "sense open". Multiply each prior by how well that door explains the reading:

$$\text{open: } 0.6 \times 0.5 = 0.3$$

$$\text{closed: } 0.2 \times 0.5 = 0.1$$

The two products are not yet probabilities, because they add to 0.4, not 1. Their sum is the evidence (G-718) $p(z)$, the overall chance of the reading, by the law of total probability (G-1053). Divide by it:

$$p(\text{open} \mid z_1) = 0.3/0.4 = 0.75$$

$$p(\text{closed} \mid z_1) = 0.1/0.4 = 0.25$$

Dividing by the sum is the same as multiplying by its inverse, written η (eta):

$$\eta = 1/0.4 = 2.5$$

This number is the **normaliser** (G-2382): whatever factor makes the products add to 1, the same job it does for the cut bell of the beam model. We never need $p(z)$ from anywhere else, because it is always the sum of the products (PR §2.2).

3.2 A second reading: Bayes' theorem with background knowledge

One reading leaves a 0.25 chance of a closed door. To be surer, the robot reads again and again gets "open". Now the belief must depend on both readings. Bayes' theorem still works if every term also carries the earlier readings behind the bar. With C standing for that background knowledge:

$$p(x \mid z, C) = \frac{p(z \mid x, C)\ p(x \mid C)}{p(z \mid C)}$$

This is **Bayes' rule with background knowledge** (G-2452) (PR §2.2). It holds for the same reason as the plain rule: the joint probability of x and z given C can be split two ways by the multiplication rule (G-2212), and setting the two equal gives the formula.

With x the door, $z = z_2$ and $C = z_1$, two of the terms are known:

- $p(x \mid z_1)$ is the answer of Section 3.1: the old posterior becomes the new prior;
- $p(z_2 \mid x, z_1)$ asks how likely the second reading is, given the door **and** the first reading.

Here the robot makes an assumption: **once the state of the door is known, the first reading tells nothing more about the second.** Each reading is the door plus its own independent sensor noise. So:

$$p(z_2 \mid x, z_1) = p(z_2 \mid x)$$

This is conditional independence (G-443): independence that holds **given** the state. It does not say that the two readings are independent on their own. They are not, as the robot's own numbers show:

$$p(z_2 = \text{open}) = 0.6 \times 0.5 + 0.2 \times 0.5$$

$$= 0.4$$

$$p(z_2 = \text{open} \mid z_1 = \text{open})$$

$$= 0.6 \times 0.75 + 0.2 \times 0.25 = 0.5$$

A first "open" makes a second "open" more likely (0.5 instead of 0.4), because it makes an open door more likely. Only when the door is known does that link disappear.

Second reading, "sense open". Same recipe, with 0.75 and 0.25 as the prior:

$$\text{open: } 0.6 \times 0.75 = 0.45$$

$$\text{closed: } 0.2 \times 0.25 = 0.05$$

$$\eta = 1/(0.45 + 0.05) = 2$$

$$p(\text{open} \mid z_1, z_2) = 0.9$$

$$p(\text{closed} \mid z_1, z_2) = 0.1$$

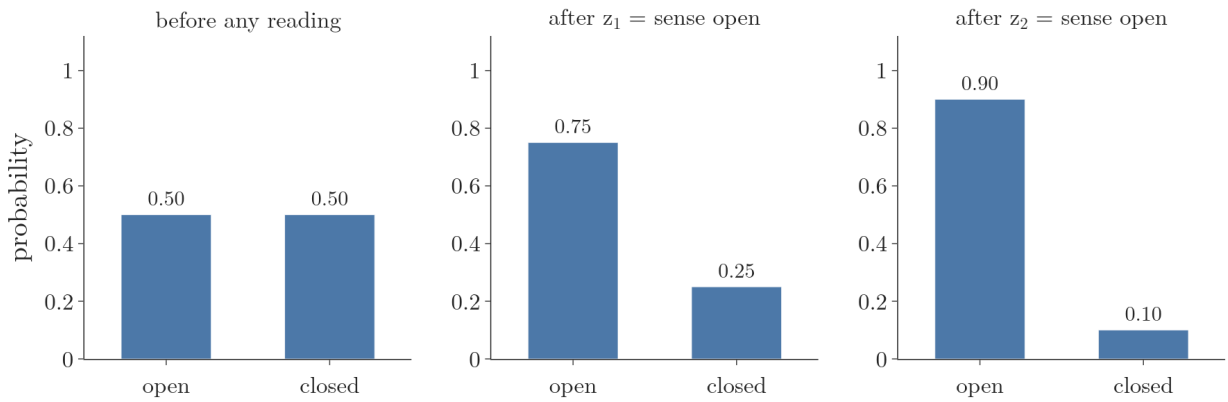


Figure 2: Belief that the door is open: 0.5 before any reading, 0.75 after the first sense open, 0.9 after the second

Figure 2 shows the belief growing with each reading. The robot never went back to the first reading: it kept one number, 0.75, and reused it. That reuse is the whole idea of a filter.

4. The Bayes filter: predict, then correct

> **Key point:** Each step has two parts. Prediction: after the command, add up every way of reaching each state. Correction: after the reading, multiply by the measurement probability and rescale.

4.1 Why a moving world needs a second step

Section 3 kept only the last answer, but it assumed nothing changes the door. Once the robot pushes, the door may change between readings, and a robot that drives changes its own position. The belief must now follow two kinds of event: commands, which change the state and usually add doubt, and readings, which add knowledge (see the predicted belief).

The honest way to know the belief is to recompute it from the whole history at every step (Figure 3, top). That is an archive, not a filter: after an hour at 10 readings a second, it holds 36,000 readings and keeps growing. The Bayes filter keeps only the last belief and updates it with the newest command and reading (Figure 3, bottom). Updating a belief from the previous belief in this way is **recursive state estimation** (G-2450) (PR §2.4).

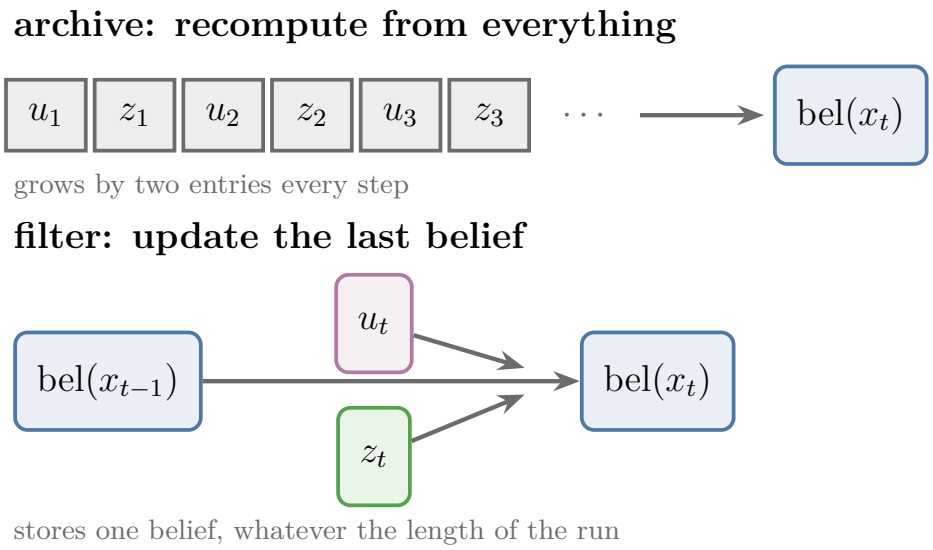


Figure 3: Top: an archive keeps every command and reading and recomputes the belief from all of them. Bottom: a filter turns the previous belief into the new one using only the newest command u_t and reading z_t

4.2 The prediction step

After the command, before the reading, the robot wants its predicted belief $\overline{\text{bel}}(x_t)$. It does not know the previous state, so it considers every previous state, asks how likely the command takes it to x_t, and weights that by how much it believed in that previous state. Adding these up is the law of total probability:

$$\overline{\text{bel}}(x_t) = \sum_{x_{t-1}} p(x_t \mid u_t, x_{t-1})$$

$$\times \text{ bel}(x_{t-1})$$

The sum runs over every possible previous state. This is the **prediction step** (G-2447), also called the motion update or time update (PR §2.4.1).

At $t = 2$, the command is "push", and the belief after $t = 1$ is 0.75 open (Section 4.3 computes it). Figure 4 draws the four paths. Two of them end with the door open:

$$\overline{\text{bel}}(\text{open}) = 1 \times 0.75 + 0.8 \times 0.25$$

$$= 0.75 + 0.2 = 0.95$$

The other two end closed:

$$\overline{\text{bel}}(\text{closed}) = 0 \times 0.75 + 0.2 \times 0.25$$

$$= 0.05$$

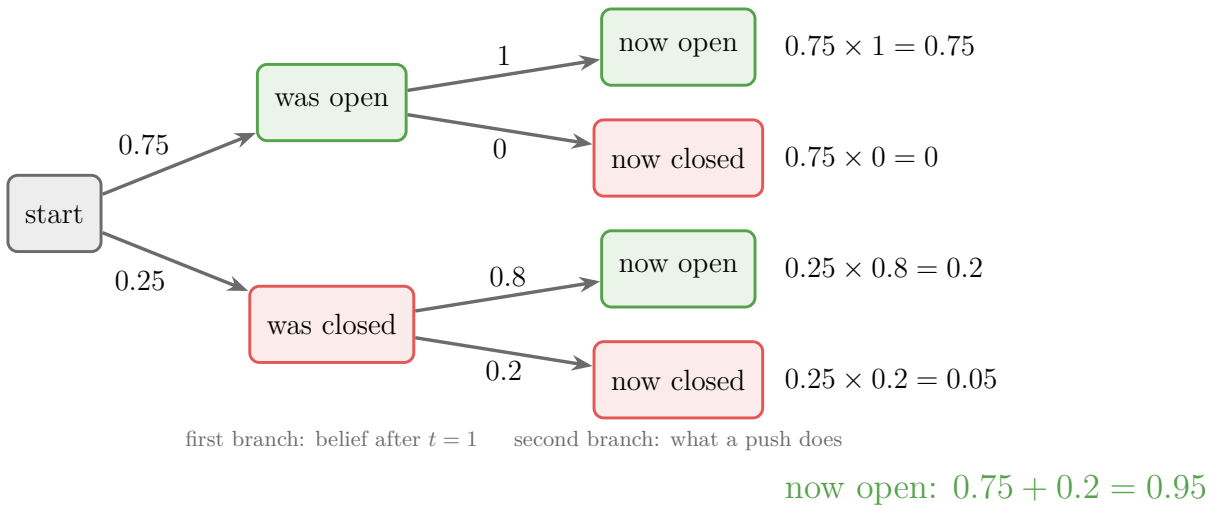


Figure 4: Probability tree for the prediction at t = 2: the door was open (0.75) or closed (0.25); a push keeps it open (1) or opens it (0.8); the two paths ending open give 0.75 + 0.2 = 0.95

The push raised the chance of an open door from 0.75 to 0.95 before the robot looked: it knows that pushing tends to open doors. At $t = 1$ the command was "do nothing", which keeps every door as it was, so the prediction changes nothing there:

$$\overline{\text{bel}}(\text{open}) = 1 \times 0.5 + 0 \times 0.5 = 0.5$$

$$\overline{\text{bel}}(\text{closed}) = 0 \times 0.5 + 1 \times 0.5 = 0.5$$

4.3 The correction step

After the reading, the robot applies Bayes' theorem of Section 3 to the predicted belief:

$$\text{bel}(x_t) = \eta \; p(z_t \mid x_t) \; \overline{\text{bel}}(x_t)$$

This is the **correction step** (G-2448), also called the measurement update (PR §2.4.1). It multiplies each state's predicted belief by how well that state explains the reading, then rescales with η so the beliefs add to 1.

At $t = 1$ the prediction changed nothing, so the correction repeats Section 3.1: the belief in open becomes 0.75.

At $t = 2$, the reading is again "sense open":

$$\text{open: } 0.6 \times 0.95 = 0.57$$

$$\text{closed: } 0.2 \times 0.05 = 0.01$$

$$\eta = 1/0.58 = 1.724$$

$$\text{bel}(x_2 = \text{open}) = 0.983$$

$$\text{bel}(x_2 = \text{closed}) = 0.017$$

These are Thrun's numbers for the same example (PR §2.4.2). Between the two steps the robot kept only 0.75 and 0.25, never the command and reading before them. Figure 5 plays the whole run.

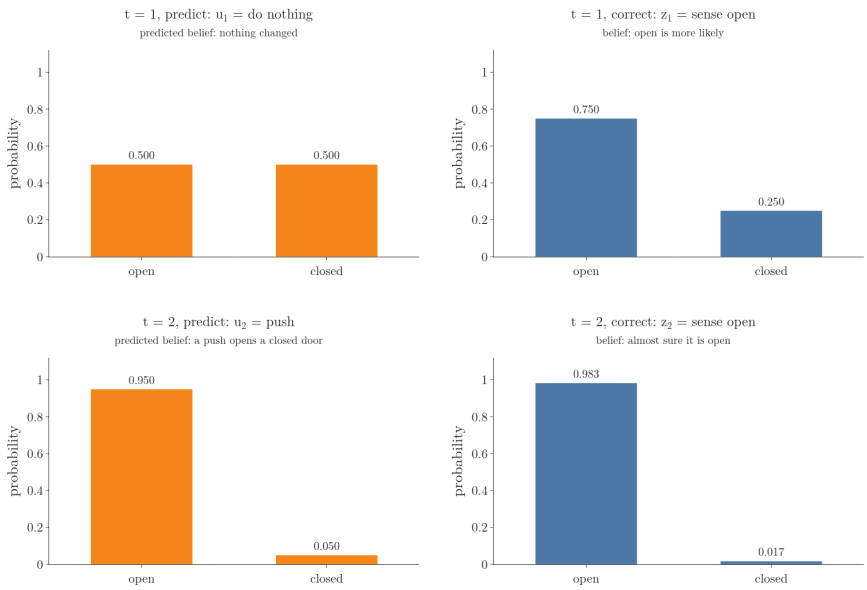


Figure 5: The door-world belief through two steps: start 0.5/0.5; predict with do nothing 0.5/0.5; correct with sense open 0.75/0.25; predict with push 0.95/0.05; correct with sense open 0.983/0.017. Predicted beliefs are orange, beliefs are blue

4.4 The algorithm

Together the two steps are the **Bayes filter** (G-2446) (PR §2.4.1). For every possible state x_t:

1. **Predict:** $\overline{\text{bel}}(x_t) = \sum_{x_{t-1}} p(x_t \mid u_t, x_{t-1})\ \text{bel}(x_{t-1})$.
2. **Correct:** $\text{bel}(x_t) = \eta\ p(z_t \mid x_t)\ \overline{\text{bel}}(x_t)$.

Then return $\text{bel}(x_t)$ and wait for the next command and reading.

> **Python:** one step of the filter for a finite set of states.

```python
import numpy as np
T_push = np.array([[1.0, 0.0],     # from open: to open, closed
                   [0.8, 0.2]])    # from closed
p_z = np.array([0.6, 0.2])         # p(sense open | open, closed)
bel = np.array([0.75, 0.25])       # belief after t = 1
bel_bar = bel @ T_push             # predict: [0.95, 0.05]
bel = p_z * bel_bar
bel = bel / bel.sum()              # correct: [0.983, 0.017]
```

The matrix product `bel @ T_push` is the sum of the prediction step: entry j adds belief times transition probability over every previous state i. The notebook `RO-014-bayes-filter.ipynb` runs both steps and checks every number above.

5. Why the two steps give the exact belief

> **Key point:** Starting from the definition of the belief, five steps lead to the filter: Bayes' rule, a Markov assumption for readings, the law of total probability, a Markov assumption for motion, and the assumption that the newest command says nothing about the past.

The two steps threw the history away. Here we show that nothing was lost: they give exactly $p(x_t \mid z_{1:t}, u_{1:t})$, the belief as defined over the whole history, and we name every assumption on the way (PR §2.4.3). The notation $z_{1:t}$ means all readings $z_1, \ldots, z_t$, and $u_{1:t}$ all commands.

Line 0, the definition. Nothing is assumed:

$$\mathrm{bel}(x_t) = p(x_t \mid z_{1:t},\ u_{1:t})$$

Line 1, Bayes' rule with background knowledge. Split the newest reading z_t off from the rest; everything older is the background C of Section 3.2. The denominator does not depend on x_t, so it becomes the normaliser η:

$$\mathrm{bel}(x_t) = \eta\ p(z_t \mid x_t,\ z_{1:t-1},\ u_{1:t})$$

$$\times\ p(x_t \mid z_{1:t-1},\ u_{1:t})$$

This line is exact; it is a rule of probability.

Line 2, the Markov assumption for readings. If the state x_t is known, the past readings and commands tell nothing more about the current reading. This is the same assumption as the door readings in Section 3.2, and the arrow structure of the dynamic Bayes network:

$$p(z_t \mid x_t,\ z_{1:t-1},\ u_{1:t}) = p(z_t \mid x_t)$$

The second factor of Line 1 is the predicted belief, by its definition. So:

$$\mathrm{bel}(x_t) = \eta\ p(z_t \mid x_t)\ \overline{\mathrm{bel}}(x_t)$$

That is the correction step. One assumption has paid for it: a history that grows forever dropped out of the measurement term.

Line 3, the law of total probability. We now unpack the predicted belief. We do not know where the robot was at $t-1$, so we bring in every possible previous state and weight each by how likely it was. This introduces x_{t-1} on purpose, to reach the previous belief:

$$\overline{\mathrm{bel}}(x_t) = \sum_{x_{t-1}} p(x_t \mid x_{t-1},\ z_{1:t-1},\ u_{1:t})$$

$$\times\ p(x_{t-1} \mid z_{1:t-1},\ u_{1:t})$$

This line is exact.

Line 4, the Markov assumption for motion. If we know the previous state, the older readings and commands tell nothing more about where the newest command takes the robot; only u_t still matters:

$$p(x_t \mid x_{t-1},\ z_{1:t-1},\ u_{1:t}) = p(x_t \mid u_t,\ x_{t-1})$$

Line 5, the newest command does not tell us the past. The last factor of Line 3 is the belief about x_{t-1}, except that it also knows u_t, the command given after time $t-1$. We assume a command chosen now says nothing about where the robot was:

$$p(x_{t-1} \mid z_{1:t-1},\ u_{1:t}) = p(x_{t-1} \mid z_{1:t-1},\ u_{1:t-1})$$

$$= \mathrm{bel}(x_{t-1})$$

This assumption can fail. A robot with collision avoidance never commands "drive 20 cm forward" while standing in front of a wall, so that command does tell us the robot was not at the wall. In most cases the command carries no such hint, and we drop it (PR §2.4.3).

The result. Putting Lines 4 and 5 into Line 3:

$$\overline{\mathrm{bel}}(x_t) = \sum_{x_{t-1}} p(x_t \mid u_t,\ x_{t-1})$$

$$\times\ \mathrm{bel}(x_{t-1})$$

That is the prediction step, and its last factor is the belief one step earlier: the formula refers to itself. That is why the robot only has to keep its last belief. The derivation used three assumptions:

1. readings depend only on the current state (Line 2);
2. the next state depends only on the previous state and the newest command (Line 4);
3. the newest command does not tell us about the previous state (Line 5).

The first two together are the Markov property (PR §2.4.4): the current state is a complete summary of the past. When they hold, the Bayes filter is not an approximation; it gives the exact belief. When they fail, for example when people walk around a robot whose map does not include them, the readings are no longer independent given the robot's pose, and the filter can become overconfident (PR §2.4.4; Fox 1999).

6. The same two steps on a continuous position

> **Key point:** For a position that can take any value, the sum of the prediction step becomes an integral; the correction step is unchanged.

Positions in a real hallway are not 2 door states but any number of metres; the filter still needs only the last belief and the newest command and reading. The belief becomes a probability density (G-1569), and the sum over previous states becomes an integral (G-957), a sum over infinitely many thin slices:

$$\overline{\text{bel}}(x_t) = \int p(x_t \mid u_t,\ x_{t-1}) \times \text{bel}(x_{t-1})\ dx_{t-1}$$

The correction step is unchanged: multiply by $p(z_t \mid x_t)$ and rescale so the area under the curve is 1 (PR §2.4.1).

The numbers for Gaussian beliefs are worked in why multiplying two bells gives a narrower bell.

7. The belief is the probabilistic information state

> **Key point:** The belief can be updated from the previous belief, the newest command and the newest reading alone. So it carries everything the history knew, and the robot can throw the history away.

Why may the robot throw its history away? The set of possible states also had two update rules, one per event. LaValle puts the two summaries side by side (Figure 6):

- after a move, the **union** of all reachable states becomes the **sum** over all previous states (the prediction step);
- after a reading, the **intersection** with the states that fit becomes **multiply by** $p(z \mid x)$ **and rescale** (the correction step).

In LaValle's words, marginalisation and Bayes' rule are "the probabilistic equivalents of union and intersection" (LaValle §11.2.3).

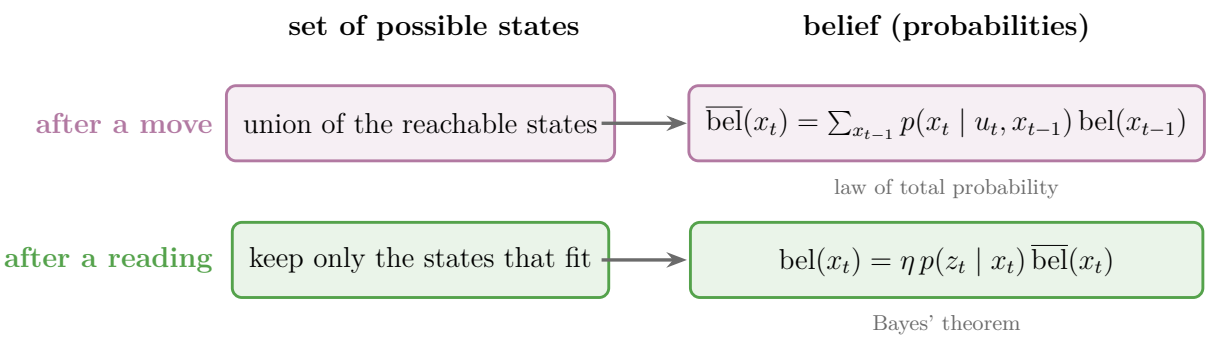


Figure 6: Left: the set-valued updates, union after a move and keep-the-states-that-fit after a reading. Right: the belief updates, the sum of the prediction step and the multiply-and-rescale of the correction step

The belief, seen as a summary of the history, is the **probabilistic information state** (G-2451) (LaValle §11.2.3). Section 5 showed that the new belief needs only the old belief, u_t and z_t. So the belief is a sufficient summary: any decision the robot could base on the whole history, it can base on the belief instead, without keeping the history (LaValle §11.2.3).

8. One equation, many filters

> **Key point:** Every filter in the rest of this chapter is the Bayes filter with a particular way of storing the belief.

The Bayes filter keeps only the last belief, but it says what to compute, not how to store that belief over a continuous pose, where the prediction sums over infinitely many previous states. Each practical filter answers that differently (PR §2.5):

- **grid filters:** chop the space into cells and keep one probability per cell, as in the hallway; the sum stays a sum;
- **particle filter:** keep a cloud of sample states; dense clouds mean high belief;
- **Kalman filter:** assume every belief is a Gaussian and keep only its mean and spread;
- **extended Kalman filter:** the Kalman filter for curved motion and sensor models.

Each is a different bet about what shape the belief is allowed to have. Knowing the Bayes filter tells us what each of them is trying to compute, and what breaks when its bet is wrong.

9. Summary

Step	When	Formula	Effect on the belief
Prediction	after the command u_t	$\overline{\text{bel}}(x_t) = \sum p(x_t \mid u_t, x_{t-1})\ \text{bel}(x_{t-1})$	shifts it and usually widens it
Correction	after the reading z_t	$\text{bel}(x_t) = \eta\ p(z_t \mid x_t)\ \overline{\text{bel}}(x_t)$	multiplies up the states that fit, then rescales

- The robot keeps only its last belief and updates it with the newest command and reading, so it never stores the history (36,000 readings after an hour at 10 a second): in the door world two steps took the belief in an open door from 0.5 to 0.983, with only 0.75 and 0.25 kept in between.
- Bayes' theorem turns the measurement probability $p(z \mid x)$, which we can measure, into $p(x \mid z)$, which the robot needs, so one reading moves the door belief from 0.5 to 0.75.
- With background knowledge, the old posterior becomes the new prior, so a second reading moves it to 0.9 without revisiting the first; this needs the readings to be independent given the state, not independent outright.
- The prediction step adds up every way of reaching each state (the law of total probability), because the robot does not know its previous state; a push raised the door belief to 0.95 before the robot looked.
- The correction step multiplies by $p(z_t \mid x_t)$ and divides by the sum, so the beliefs add to 1; the normaliser never has to be known in advance.
- The derivation is exact under three assumptions (readings depend only on the current state, motion only on the previous state and command, the newest command says nothing about the past), so the filter fails in the ways those assumptions fail.
- On a continuous position the sum of the prediction step becomes an integral, because there are infinitely many previous positions to add up; the correction step is unchanged.
- The new belief needs only the last belief, u_t and z_t, so it carries everything the history knew: it is the probabilistic information state.

The robot can now keep its belief up to date from the newest command and reading alone. Still missing: the prediction sums over every previous state, and a position has infinitely many, so the sum cannot be done as written; grid filters turn it into a sum over 10 cells.

10. Sources

Built from

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11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Bayes' rule with background knowledge (G-2452)	Bayes' theorem with an extra condition C in every term, $p(x \mid z, C) = p(z \mid x, C)\ p(x \mid C)/p(z \mid C)$, such as earlier readings; it lets the posterior after one reading serve as the prior for the next.
Recursive state estimation (G-2450)	Estimating the state by updating the previous estimate with only the newest command and reading, instead of recomputing from the whole history; it keeps the work per step fixed however long the robot runs.

Term	Meaning
Prediction step (motion update) (G-2447)	The Bayes-filter step after a command: for each state, add up the probability of reaching it from every previous state, weighted by the belief in that state, $\overline{\text{bel}}(x_t) = \sum p(x_t \mid u_t, x_{t-1})\ \text{bel}(x_{t-1})$; it moves the belief with the robot and usually widens it, because the motion is uncertain; a near-certain command, such as a push that opens a door, can make the belief sharper.
Correction step (measurement update) (G-2448)	The Bayes-filter step after a reading: multiply each state's predicted belief by the measurement probability $p(z_t \mid x_t)$ and rescale so the beliefs add to 1; it raises the states that explain the reading and sharpens the belief.
Bayes filter (G-2446)	The algorithm that keeps a robot's belief up to date one step at a time: a prediction step after each command and a correction step after each reading; it gives the exact belief without storing the history, under the Markov assumption.
Probabilistic information state (G-2451)	The belief viewed as a summary of the robot's history: a probability distribution over states that can be updated from itself, the newest command and the newest reading, so it carries everything the history knew and the history can be discarded.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.
Evidence (G-718)	The overall chance of seeing what we observed, counted over every possible hypothesis, written $P(E)$. Bayes' theorem divides by it: posterior = likelihood × prior / evidence.
Law of total probability (G-1053)	The rule for the overall probability of an event: add its probability under each case, weighted by how likely that case is, $P(B) = \sum_i P(B \mid A_i)P(A_i)$, when the cases A_i are mutually exclusive and cover every possibility.

Term	Meaning
Normalizer η (G-2382)	The factor that rescales a set of numbers, or a density, so that they add up (or integrate) to 1; it is always one over the current total: the sum of the products in the Bayes filter ($1/0.4 = 2.5$ after a reading in the door world), the area under the cut curve in the beam model ($1/0.655 = 1.53$).
Multiplication rule (G-2212)	The chance that both A and B happen is the chance of B times the chance of A given B: $P(A \cap B) = P(A \mid B) \times P(B)$.
Conditional independence (G-443)	Independence that holds once a third variable is known: given the class, knowing one feature tells nothing more about another. Naive Bayes assumes it so it can multiply one probability per feature.
Probability density (G-1569)	The height of a continuous distribution's curve; compares how likely nearby values are.
Integration (G-957)	Finding the area under a curve by adding up infinitely many thin strips.
Gaussian distribution (G-827)	Another name for the normal distribution: the symmetric, bell-shaped continuous distribution set by its mean and standard deviation, used to model many measurements.